Simone Rea

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Education

Alma Mater Studiorum – University of Bologna — **Bologna, Italy**
Bachelor's Degree in Computer Engineering — *2024 – Present*
- Current GPA: 29.35/30 (equivalent to **3.926/4.0**)

Eindhoven University of Technology – TU/e — **Eindhoven, Netherlands**
Incoming Student – Erasmus Program — *February 2027 – July 2027*

Yonsei University — **Seoul, South Korea**
Exchange Program — *September 2026 – December 2026*
- **Relevant Coursework:** System Programming, Quantum Computer Programming, Electronic Circuits, Deep Neural Networks, Digital Microelectronics

Istituto Tecnico-Scientifico Ettore Majorana — **Orvieto, Italy**
Traditional Scientific High School Diploma — *2019 – 2024*
- Graduated with **100/100 cum laude**

Experience

Biozentrum – University of Basel — **Basel, Switzerland**
Summer Researcher, Engel's Lab — *July 2026 – September 2026*
- Performed **expansion microscopy (ExM)** on diverse biological samples (*Chlorella*, *Chlamydomonas*, plant leaf tissue) and characterized subcellular structures via **confocal/widefield fluorescence imaging**.
- Investigated a recurring **"Fusilli Fold" structural motif** in protein sequences via a bioinformatics pipeline combining **MEME**, **MUSCLE**, **HMM**, and **Jackhmmer**.
- Implemented and adapted **CryoTIGER**, a deep-learning-based tilt-interpolation tool for cryo-electron tomography, to improve 3D reconstruction quality.

ADM Team – Alma Mater Studiorum — **Bologna, Italy**
Software Engineering Member — *November 2024 – November 2025*
- Developed backend services in **Go** to implement a DNS system backed by **PowerDNS**.
- Implemented and tested HTTP APIs using `curl` for request validation and debugging.
- Acquired practical knowledge of distributed **system architecture** and service-oriented design.
- Set up **Prometheus** for monitoring, metrics collection, and basic alerting.
- Managed application **deployment** and environment configuration in Linux-based systems.

Unibo Motorsport — **Bologna, Italy**
Novice Member – Driverless Department — *March 2025 – June 2025*
- Collaborated in a multidisciplinary university team building autonomous systems.
- Supported the Software Infrastructure sub-department using **C#** and **Unity** for driving simulator development.
- Designed and implemented a custom control interface for an autonomous robot using **Python** and **ROS2**.

Projects & Achievements

- **CyberChallenge Program:** Selected among the **top 25 Computer Science students** at the University of Bologna for a national cybersecurity training program.
- **High School Math Olympics:** Competed in national mathematics problem-solving competitions.

Certifications

- **IELTS Academic:** Overall Band Score 7.5

Technical Skills

Programming Languages: Python, Rust, Java, C++, C, JavaScript, Go, SQL

Web Technologies: HTML, CSS, Node.js

Tools & Software: Unity, Figma, Photoshop, Prometheus, Proxmox

Systems & Concepts: Linux, ROS2, Operating Systems, Computer Networks, POSIX, LDAP